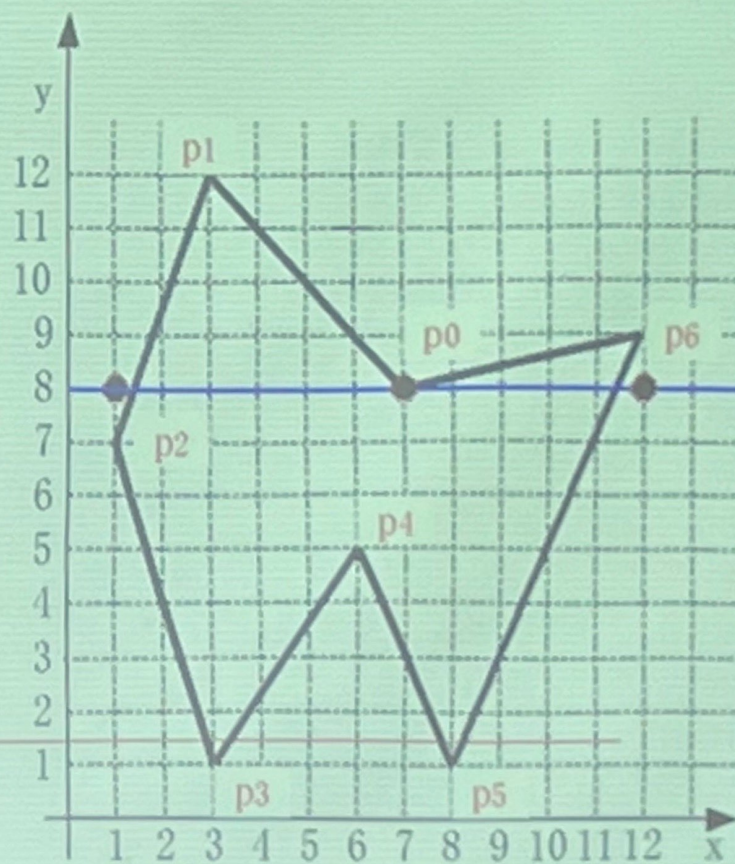


随堂作业

1、试用中点画线法扫描转换一条连接两点 $(0, 0)$ 和 $(8, 6)$ 的直线段。

2、试用Bresenham画线法扫描转换一条连接两点 $(0, 0)$ 和 $(8, 6)$ 的直线段。

3、如下图所示多边形，若采用扫描转换算法（NET边表算法）进行填充，试写出该多边形的ET表和当扫描线 $Y=2$ 时的有效边表AET表（活性边表），并说明在该处的填充区间。



$(0, 0), (8, 6)$

$\Delta x = 8, \Delta y = 6, k = \frac{\Delta y}{\Delta x} < 1$

$D_0 = \Delta x - 2\Delta y = -4, \begin{cases} d < 0 \Rightarrow y = y + 1 \\ d \geq 0 \Rightarrow y = y \end{cases}$

$2\Delta x - 2\Delta y = 4$
 $-2\Delta y = -12$

x	0	1	2	3	4	5	6	7	8
y	0	1	1	2	3	4	4	5	6
d	-4	0	-12	-8	-4	0	-12	-8	4

$(0, 0) (8, 6)$

$ax + by + c = 0.$

$\frac{3}{4} = 1$

$a = y_0 - y_1 = -6, b = x_1 - x_0 = 8$

$x_0 = 2a + b = -4$ $\left(\begin{array}{l} D < 0 \\ D > 0 \end{array} \right. \begin{array}{l} y = y + 1 \\ y = y \end{array} \right)$

$2a + 2b = 4$

$2a = -12$

x	0	1	2	3	4	5	6	7	8
y	0	1	1	2	3	4	4	5	6
d	-4	0	-12	-8	-4	0	-12	-8	-4

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改进2: 用 $E=2e^{\Delta x}$ 来替换 e , 消除浮点数

- $e_{\text{初}} = -0.5$
- 每走一步有 $e = e + k$
- if ($e > 0$) then $e = e - 1$



- $E_{\text{初}} = -0.5 * 2^{\Delta x} = \underline{-\Delta x}$
- 每走一步有 $E = (e + k) * 2^{\Delta x} = \underline{E + 2^{\Delta y}}$
- if ($e > 0$) then $E = (e - 1) * 2^{\Delta x} = \underline{E - 2^{\Delta x}}$

$$(0, 0) \quad (8, 6)$$

$$\Delta x = 8, \Delta y = 6.$$

$$2\Delta y = 12.$$

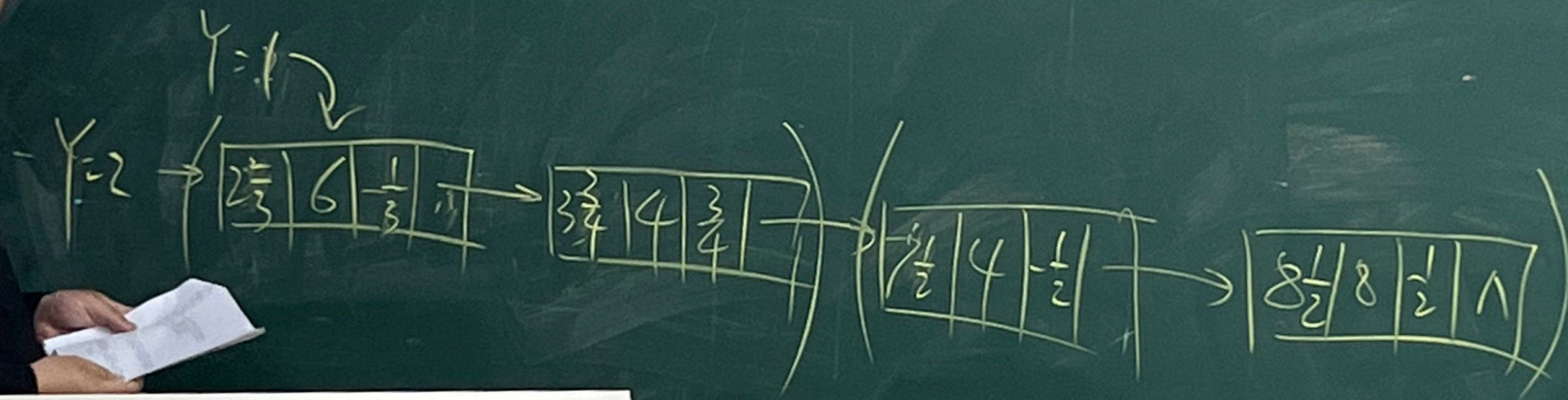
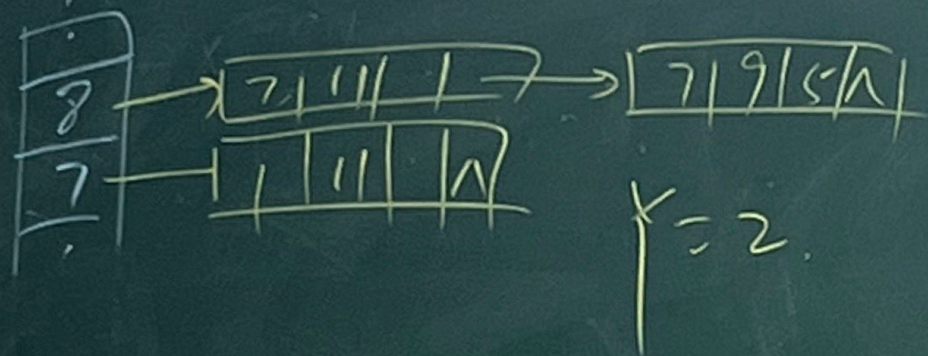
$$-2\Delta x = -16.$$

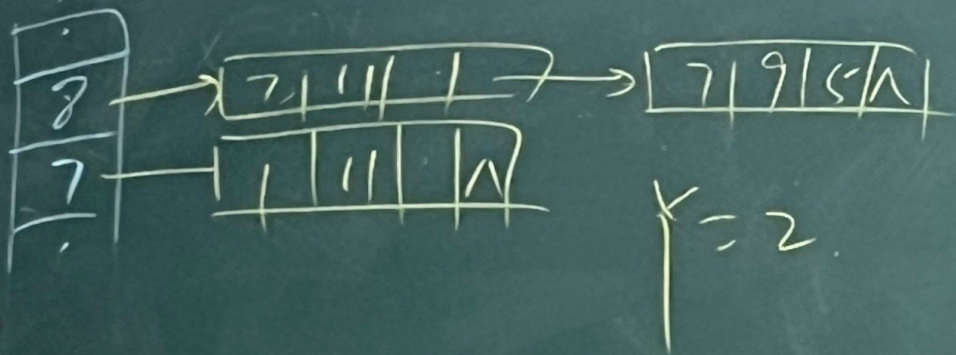
x	0	1	2	3	4	5	6	7	8
y	0	1	1	2	3	4	4	5	6
e	-8	-12	0	-4	-8	-12	0	-4	-8
e+y	4	0	12	8	4	0	12	8	4

$$E_0 = -0x$$

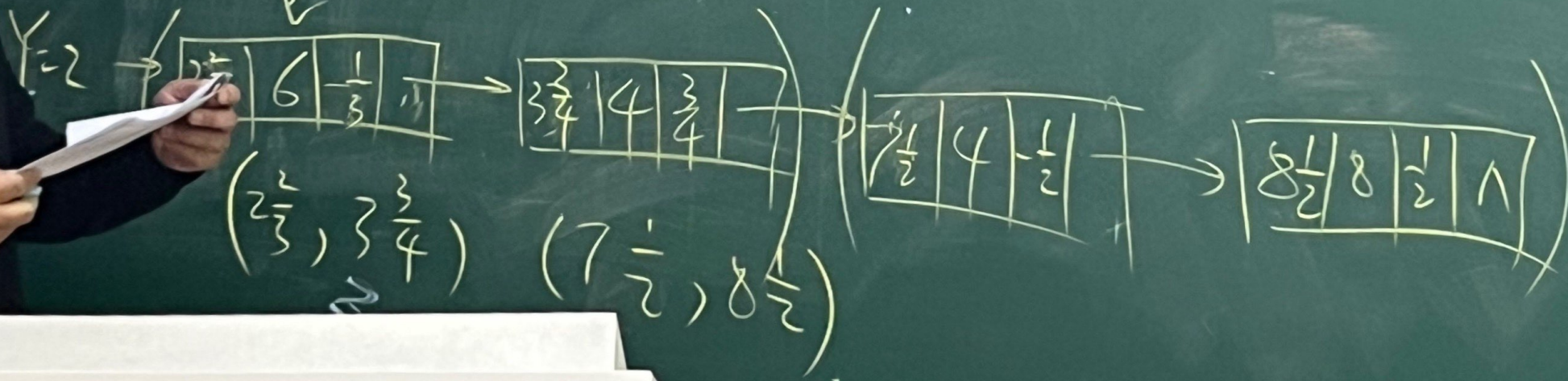
$$\Rightarrow \underline{T = E + 2\Delta y.}$$

$$\begin{cases} e > 0 & y = y+1, e = e - 2\Delta x \\ e \leq 0 & y = y. \end{cases}$$





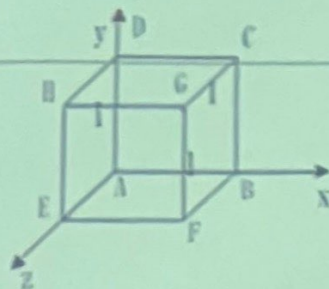
$Y=1 \rightarrow (3, 2), (8, 2)$



测试题



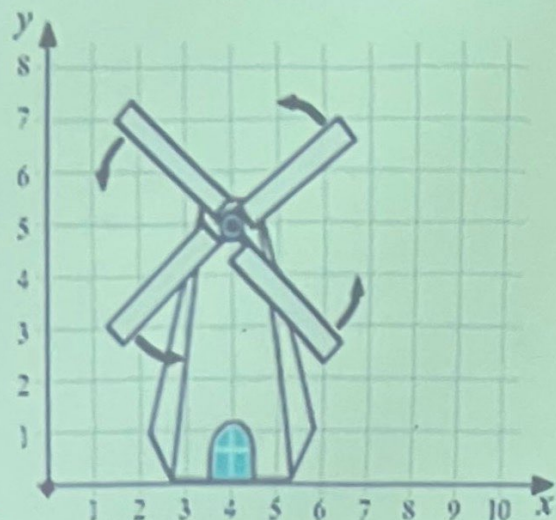
1. Generate the one point perspective of the object, where $l=0.4$, $m=0.5$, $n=2$, $d=1.5$.



2. Show that the reflection about the line $y = -x$ is equivalent to a reflection relative to the y -axis followed by an anticlockwise rotation of 90° .

3. A windmill is modeled in its final position in world coordinates as shown. Write a transformation that could be applied to just the windmill vanes (风车叶片), that will rotate them anticlockwise (逆时针) 45° so that they pivot about their axle as shown.

(请用齐次坐标表示)



One Point Perspective (一点透视图)

给出三维物体中任一点
(x, y, z) 一点透视变换
矩阵形式:

$$T_p = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{d} \\ l & m & 0 & \frac{n}{d} + 1 \end{bmatrix}$$

$$[x' \quad y' \quad z' \quad 1] = [x \quad y \quad z \quad 1] \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{d} \\ l & m & 0 & \frac{n}{d} + 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{x+l}{\frac{(n+z)}{d} + 1} & \frac{y+m}{\frac{(n+z)}{d} + 1} & 0 & 1 \end{bmatrix}$$

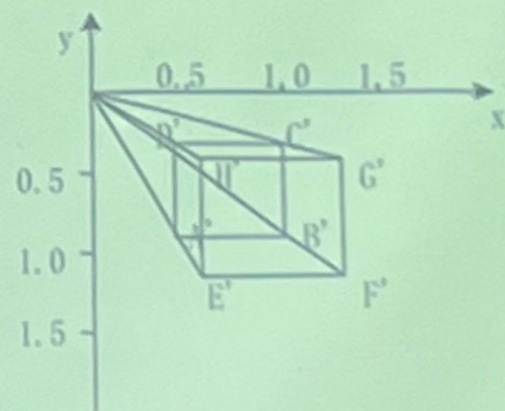
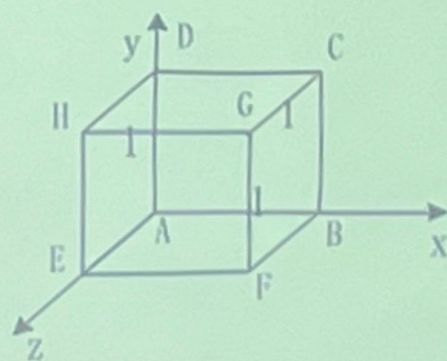
将上述值代入一点透视变换矩阵, 得到:

$$\begin{matrix} A \\ B \\ C \\ D \\ E \\ F \\ G \\ H \end{matrix} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{d} \\ l & m & 0 & \frac{n}{d} \end{bmatrix}$$

$$\begin{matrix} l = 0.8, m = -1.6, \\ n = -2, d = -2.5 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -0.4 \\ 0.8 & -1.6 & 0 & 1.8 \end{bmatrix}$$

$$= \begin{bmatrix} 0.8 & -1.6 & 0 & 1.8 \\ 1.8 & -1.6 & 0 & 1.8 \\ 1.8 & -0.6 & 0 & 1.8 \\ 0.8 & -0.6 & 0 & 1.8 \\ 0.8 & -1.6 & 0 & 1.4 \\ 1.8 & -1.6 & 0 & 1.4 \\ 1.8 & -0.6 & 0 & 1.4 \\ 0.8 & -0.6 & 0 & 1.4 \end{bmatrix} = \begin{bmatrix} 0.44 & -0.89 & 0 & 1 \\ 1 & -0.89 & 0 & 1 \\ 1 & -0.33 & 0 & 1 \\ 0.44 & -0.33 & 0 & 1 \\ 0.57 & -1.14 & 0 & 1 \\ 1.29 & -1.14 & 0 & 1 \\ 1.29 & -0.43 & 0 & 1 \\ 0.57 & -0.43 & 0 & 1 \end{bmatrix} \begin{matrix} A' \\ B' \\ C' \\ D' \\ E' \\ F' \\ G' \\ H' \end{matrix}$$



单位立方体的一点透视

One Point Perspective (一点透视图)

一点透视

平移

$$T_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ l & m & n & 1 \end{bmatrix}$$

透视

$$T_p = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_p = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

正投影

$$T_{p1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1/d \\ l & m & 0 & n/d + 1 \end{bmatrix}$$

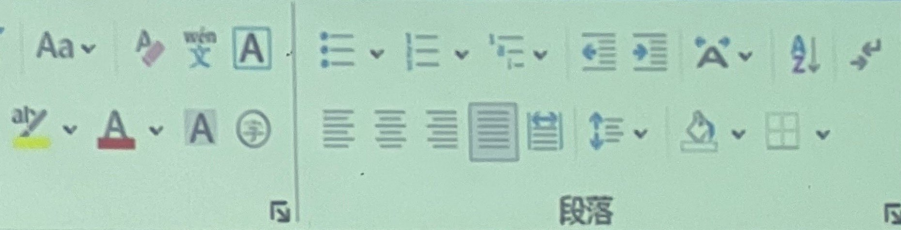
最终矩阵



$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \overset{0}{\cancel{\cos 90}} & \overset{1}{\cancel{\sin 90}} & 1 \\ \overset{0}{\cancel{\sin 90}} & \overset{0}{\cancel{\cos 90}} & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

=

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Grade

C) May use 4-connected or 8-connected approach

D) Must not use recursive approach

←

4. The modern PC graphics card consists of three main components: ().

A) GPU

B) video memory

C) RAMDAC

D) ROM BIOS

←

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5. Properties of good line drawing algorithm ().

A) Line should appear straight

B) Line should be drawn rapidly

C) Lines should terminate accurately

D) Lines should have constant density

←

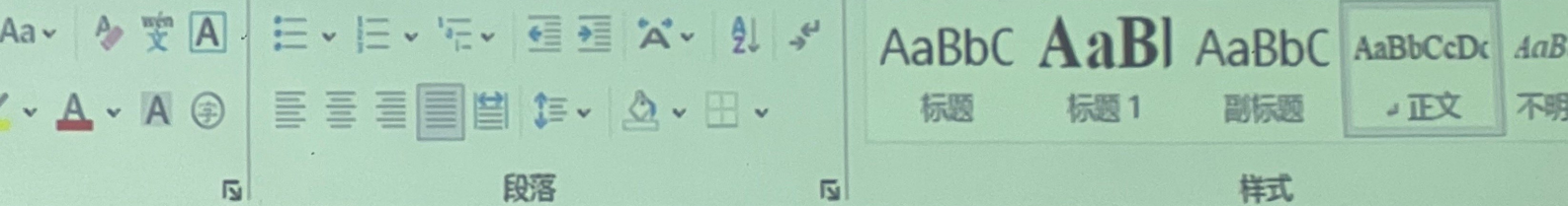
Course Name Page 1 of 5 pages

II. Blank Fillings (5 items, 2 points each)

辅助功能: 不可用



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II. Blank Fillings (5 items, 2 points each)

Score

1. In perspective projection, the maximum number of major vanishing points is _____.

2. X 扫描线算法中, 每次用一条扫描线进行填充, 对一条扫描线填充的过程可分为 4 个步骤: _____、_____、_____、_____。

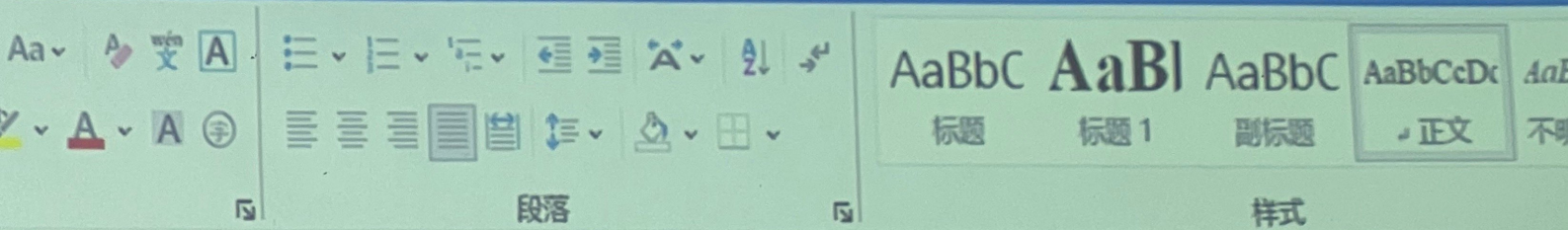
3. 对于基本几何变换, 一般有平移、旋转、对称和错切等, 这些基本几何变换都是相对于_____和_____进行的几何变换。

4. 具有 256 级灰度、分辨率为 1024×1024 个像素阵列的光栅扫描式显示器需要_____KB 的缓冲器。

5. 利用中点画线法扫描转换直线 $P(0,0)$ 和 $P(5,2)$ 时, 最大的步长方向为: _____。

III. Judgement (10 items, 1 point each)

Score



1. Both Boundary Filling and Flood filling algorithms are non-recursive techniques.

()

2. To modify the object shape, shearing transformation cannot be used. ()

3. Using Cohen-Sutherland line clipping, it is impossible for a vertex to be labeled

1111.

()

4. Translation moves objects without deformation.

()

5. Sutherland-Hodgeman clipping algorithm clips any polygon against a concave clip polygon.

()

6. 字符的图形表示分为点阵和矢量两种形式。

()

7. 视区大小不变, 当窗口变大时, 映射到视区的物体也会相应变大。 ()

8. 使用齐次坐标可以将 n 维空间的一个点向量唯一映射到 $n+1$ 维空间中。()

9. 反走样技术可以完全消除走样现象。 ()

10. LCD 表示发光二极管显示器。 ()

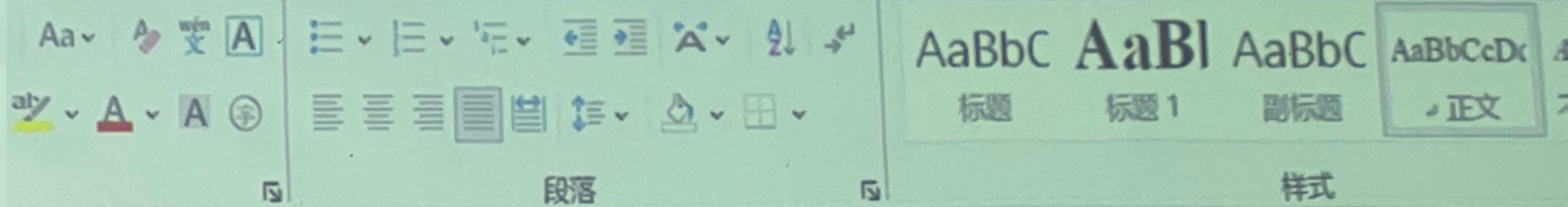
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辅助功能: 不可用

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IV. Short Answer Questions (5 items, 4 points each)

Score

1. What is the difference between computer graphics and image processing?

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2. What do you mean by projection? Describe the difference between parallel projection and perspective projection.

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3. Explain what is aliasing and what is antialiasing? Please list three antialiasing techniques.

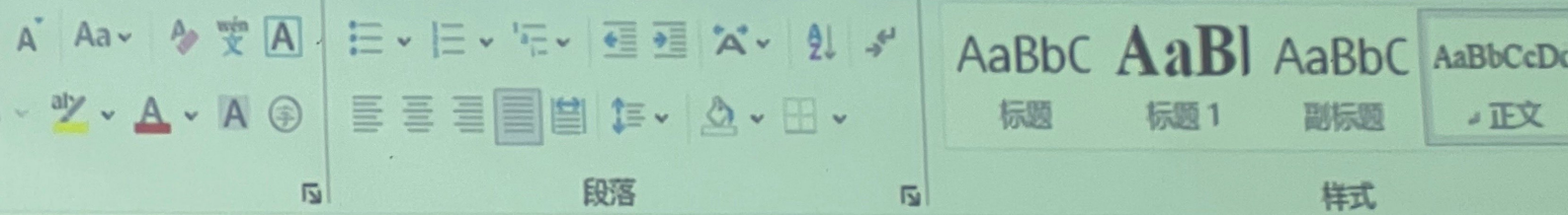
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4. Describe the 2D rotation transformation process with respect to any reference point.

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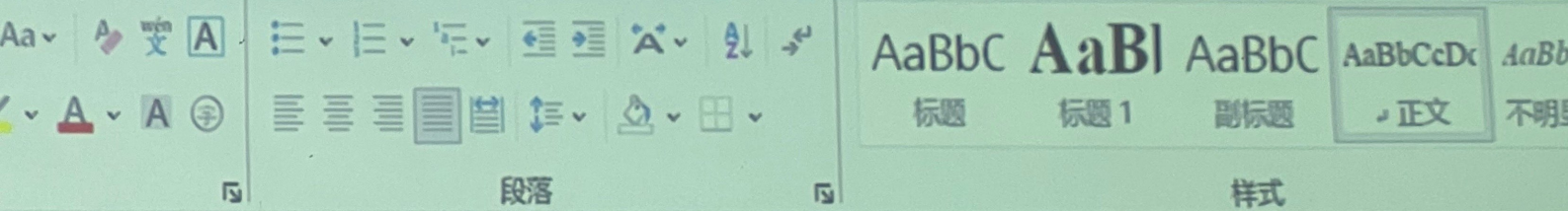


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5. 描述改进的一个深度缓存变量的 Z-buffer 算法的流程。←

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V. Algorithms and Calculations (5 items, 10 points each)

Score

1. Find the transformation matrix that reflects a polygon whose vertices are $A(-1, 0)$, $B(0, -2)$, $C(1, 0)$, and $D(0, 2)$ about the Y-axis, and write the normalized homogeneous coordinates of each vertex of the polygon after transformation.

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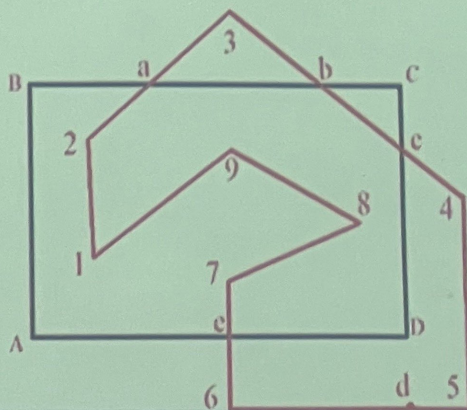
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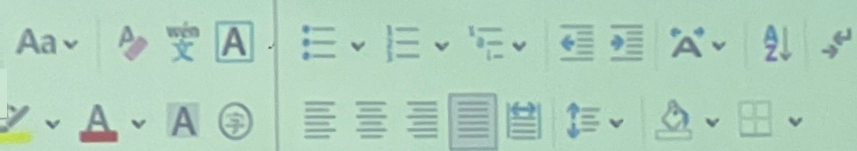
2. Clip the polygon using the Sutherland-Hodgeman algorithm. The Input is 123456789.

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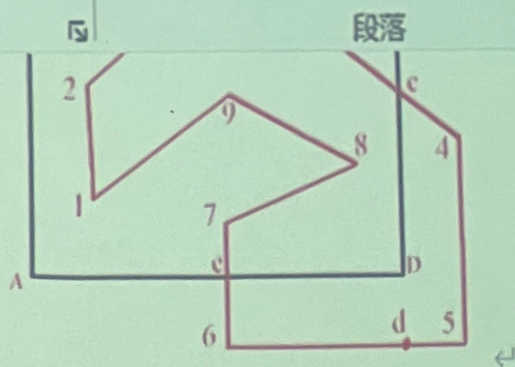
辅助功能: 不可用

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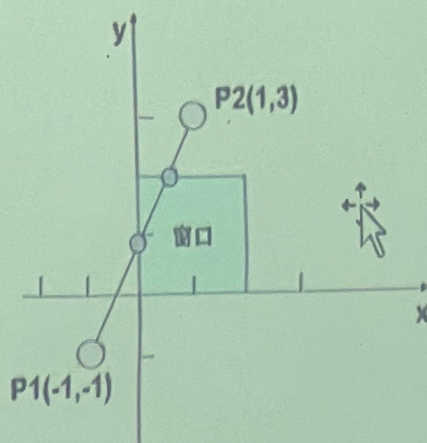


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标题 标题 1 副标题 正文 不明

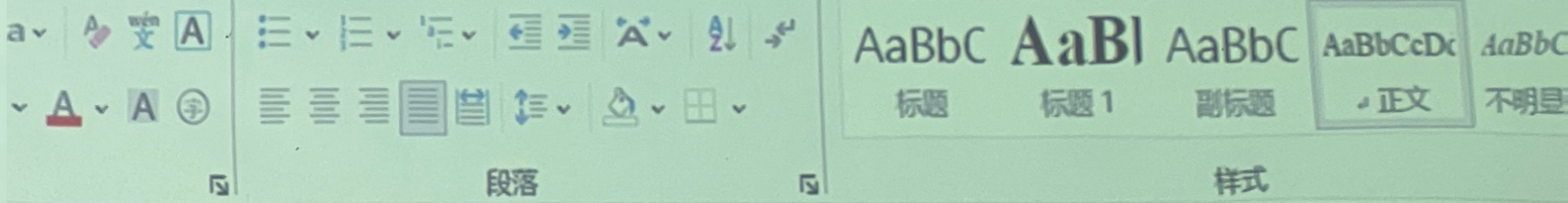
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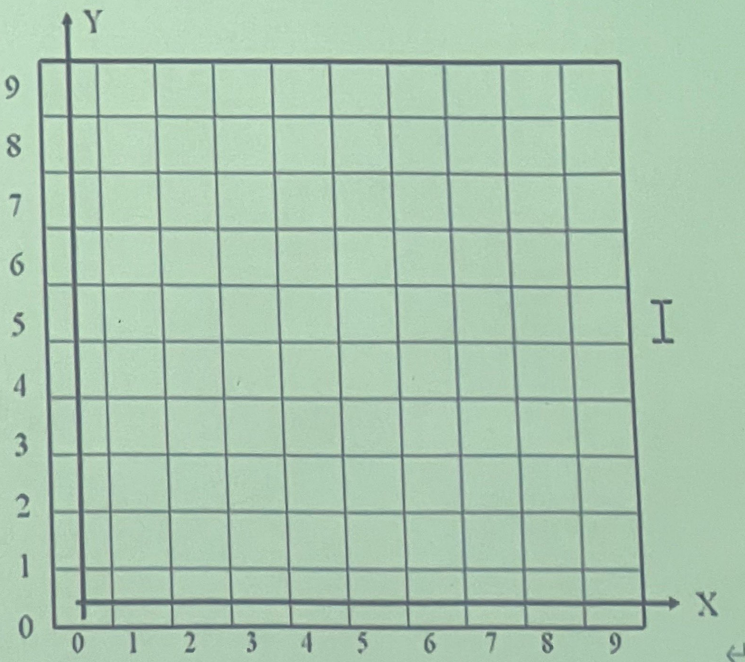
3. Use the Liang-Barsky algorithm to clip the line P_1P_2 . The four vertex coordinates of the window are $(0, 0)$, $(0, 2)$, $(2, 0)$ and $(2, 2)$.



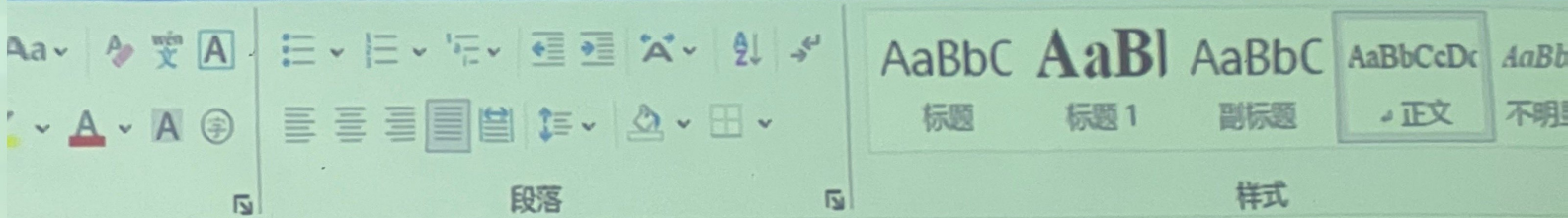
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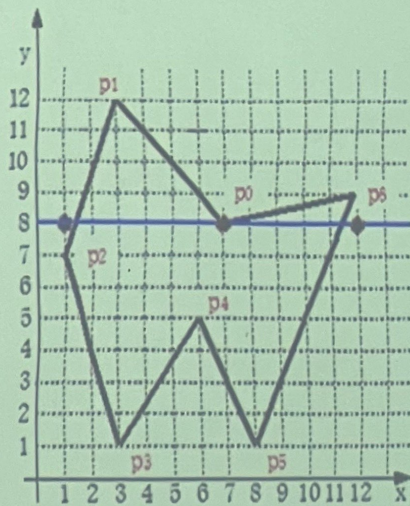
4. 用 DDA 算法生成 $A(2, 2)$, $B(8, 4)$ 线段, 请画出其轨迹, 并注明坐标, 并写出计算过程。



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5. 采用改进的有效边表算法对下图所示多边形进行填充, 试写出该多边形的 ET 表(也称为 NET 表)和当扫描线 $Y=4$ 时的 AET 表。←



(a) 多边形 $P_0P_1P_2P_3P_4P_5P_6P_0$ ←